

Project Initialization and Planning Phase

Date	13 July 2026
Project Title	Cosmetic Insights : Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau** project aims to analyze cosmetic product data using interactive Tableau dashboards and a Flask web application. The project transforms raw cosmetic data into meaningful visualizations, enabling users to compare brands, prices, ratings, and skin type suitability. This solution improves decision-making by providing clear, interactive, and data-driven insights.

Project Overview	
Objective	The primary objective of this project is to analyze cosmetic product data and develop interactive Tableau dashboards that help users compare products based on brand, category, price, rating, and skin type. The project also integrates the dashboards into a Flask web application for easy online access.
Scope	The project includes collecting and preparing cosmetic product data, creating interactive dashboards and stories in Tableau, publishing dashboards on Tableau Public, integrating them into a Flask web application, and providing users with meaningful insights through interactive visualizations.
Problem Statement	
Description	Consumers often find it difficult to compare cosmetic products because product information is spread across multiple sources. Existing platforms provide limited visualization and comparison features, making it challenging to identify the best products based on price, ratings, brand, and skin type.

Impact	Solving these challenges enables users to make informed purchasing decisions, improves product comparison, provides better market insights, and enhances the overall user experience through interactive dashboards.
Proposed Solution	
Approach	Develop an interactive Tableau dashboard that analyzes cosmetic products and displays KPIs, charts, filters, and stories. The dashboards are published on Tableau Public and integrated into a Flask web application, allowing users to access insights from anywhere.
Key Features	• Interactive Tableau dashboards • KPI cards for Total Products, Brands, Average Price, and Average Rating • Product comparison by brand, category, price, and rating • Skin type analysis (Dry, Oily, Combination, Normal, Sensitive) • Interactive filters and calculated fields • Tableau Story for sequential insights • Responsive Flask web application • Tableau Public deployment

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 or higher
Memory	RAM	8 GB
Storage	Disk space for data, models, and logs	20 GB or more
Software		

Visualization Tool	Dashboard Development	Tableau Desktop
Backend Framework	Web Application	Python (Flask)
Frontend	User Interface	HTML, CSS, JavaScript, Bootstrap
Development Environment	IDE	Visual Studio Code / PyCharm
Data		
Dataset	Cosmetic Product Dataset	CSV Format
Dashboard Hosting	Tableau Public	Published Dashboard
Web Hosting	Vercel	Flask Web Application
Version Control	GitHub	Source Code Repository